Accent-Aware Streaming Multilingual Speech-to-Speech Translation with Confidence-Guided Incremental Decoding

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Abstract- The speech-to-speech (S2ST) translation literature is largely focused on monolingual, non-streaming, and speech-to-text-to-speech translations. We present an accent-aware streaming S2ST system as a working prototype. Our implementation is a Python framework able to perform S2ST in English-Hindi-Spanish. The framework takes speech at 16-kHz, extracts MFCCs, performs English speech recognition with an accent proxy, which is then passed to a generic speech recognizer. It employs a stability gate prior to releasing the recognition results. It also utilizes a confidence gate. The framework implements eight conversational actions in Spanish and Hindi and uses eSpeak for synthesizing speech. For evaluation, we employed a held-out method and fixed the window size to 1024 samples. The generic-speech stability gate was used to control latency. English-to-Hindi and English-to-Spanish BLEU scores improved from 78.66 to 88.88 and 83.52 to 90.88, respectively. Our system employs several techniques which enable it to evaluate accent sensitivity, confidence, stability, and latency. These techniques are evaluated end-to-end and in the context of S2ST, including the synthesis of target speech. A synthetic accent proxy is not intended to evaluate Indian accents.

Keywords—accent adaptation, multilingual ASR, simultaneous speech translation, speech synthesis, streaming latency, low-resource speech.

I. INTRODUCTION

Speech-to-Speech Translation (S2ST) works best when the user can communicate in a natural, fluid way. Users also need to trust that the translation provided is accurate and may not be corrected in subsequent audio streams. If the system has to wait to translate a complete user utterance, the interaction will be unnecessarily slow. This study assumes that releasing or committing to a recognition decision can also be modeled as a conscious choice.

Existing research has developed effective multilingual speech recognition and translation systems as well as direct speech-to-speech translation [1], [2], [3], [4], [5]. Their strengths in performing tasks may lead some to believe that their effectiveness in a streaming context will also be robust. However, that may not be the case. This study evaluates a matched-ablation prototype that integrates confidence-based recognition routing, stability-gated commitment, fixed-phrase translation, and local speech synthesis, with measurements of recognition error and first-audio latency.

This method uses a combined, end-to-end decision and release model for the two phases of the S2ST cascade. For English utterances, a router compares recognition costs, which are stored in a memory bank and an accent-proxy memory bank. The router selects the branch with the more favorable recognition cost. The other branch is blocked until the decision cost is less than a threshold value. A second block occurs when a partial hypothesis is not in agreement with the previous one. Speech is output when the speech synthesizer is confident in both partial hypotheses. Spoken output is considered irreversible. The model is evaluated against single-component baselines to isolate effects.

Rather than present a blueprint, the authors tested their idea using artificial speech. They evaluated their idea in comparison to fixed-window streaming. The authors observed improvements in the median English WER and median latency. The accent-proxy English subgroup showed a larger WER reduction, whereas generic English WER increased with routing. Pooled median first-audio latency declined, but p95 latency rose. The accent-proxy tests evaluate controlled TTS-rendered speech rather than naturally spoken Indian English; perceived speech quality was not assessed.

II. RELATED WORK AND RESEARCH GAP

A. Accent Robustness and Multilingual Speech Coverage

Accuracy in ASR is influenced by a number of factors, including the speaker's accent and recording environment. The focus of Svarah is the assessment of ASR for Indian accents. FLEURS provides coverage for a range of multilingual speech, and both resources encourage assessment of ASR for accents other than those predominant in training datasets [5], [6]. Aggregate scores for the resources, however, cannot be interpreted as scores for the ASR prototype. Svarah and FLEURS employ general and accent-related templates, respectively, to create benchmarks for evaluation. The prototype freezes the implementations for the template banks and uses them to evaluate routing in ASR systems.

State-of-the-art Speech Recognition (SR) systems, including Whisper [4] and XLS-R [1], provide end-to-end (E2E) systems for SR. In this work, a lightweight MFCC-based template recognizer is used to provide an a priori policy and routing in a Speech-to-Text (STT) system. The E2E recognizer systems are not developed, and thus, the focus of this work is not in the area of new research on Automatic Speech Recognition (ASR). Systems with template-based recognizers in different policy branches utilize the same setting of feature space and phrase dictionary. Only the template banks differ. The template banks are adapted to specific accents. Consequently, in this work, policy and routing differences are evaluated for different template banks.

Development and test data for ASR systems are created using different voice settings. Development data are used to set the threshold for routing and confidence. Test data are played in all conditions. The test data in this work are rendered using a Hindi Text-to-Speech (TTS) engine and a controlled acoustic proxy for the range of accents found in Indian English. Findings from this work should not be generalized for all human accents.

The resulting research gap is narrower and testable: the study looks at the integration of speech output and translation in this paper with accent-sensitive branch selection and confidence-based incremental stability to further improve real-time speech-to-speech (S2ST) systems. The focus is on routing errors, release stability,

latency, and shortcomings in held-out evaluations. The other features of the system are designed to allow negative results to be observed, particularly false accent routing. This is the case even if parameters are fixed, inputs are held out, and outputs are measured.

B. Translation, Streaming Policies, and Output Speech

Multilingual S2ST systems (i.e. systems that convert speech to speech) and corpora (i.e. collections of S2ST data) as well as speech translation systems (e.g. CVSS) and speech unit-based translation systems provide evidence in support of cross-language speech translation [2], [3], [7]. Concurrent translation research concerns the effects of different policies and mechanisms for inter-process communication on latency [8], [9], [10]. In this research, the focal point is not on developing a new speech translation model or on provocative speculation (e.g. accent routing) about speech translation models. The objective is to produce an integrated system that comprises, in a research prototype implementation, all the essential components (e.g. speech recognition, speech translation, text to speech, speech to speech, and timing of first audio) for speech translation, and to evaluate the integrated system in the context of a speech-to-speech transmission service.

A fixed-window speech translation system can provide real-time speech-to-speech translation. However, such a system is not very reliable. If the system takes more time to process an utterance, it will be able to provide more reliable speech translation. The system presented in this research uses a technique called a stability gate to increase the reliability of the speech translation. The system provides an estimate of the speech translation latency and shows that the technique is able to reduce the speech translation latency.

Audible output is considered final. The prototype captures the selected branch, hypothesis, commitment time, confidence, target phrase, and the elapsed time to synthesize the output (if any). It also captures if output was generated due to deadline. Reporting this information is critical, since an incorrect final transcript cannot retract a previously spoken phrase. It reports estimated Word Error Rate (WER), and BLEU scores for phrase translation. It reports first audio latency. It reports recognition error, if any.

The phrase-level translation interface of the controlled experiment emulates the decision service. This fixed English, Spanish, and Hindi inputs and outputs. Although it restricts the command condition to a large extent, it ensures uniformity in the remaining variables in the different conditions.

Figure 1. Real-World Smartphone Speech Translation System



Source: National Institute of Standards and Technology (NIST), “Smart Phone Voice Translation System,” July 21, 2010. Image credit: NIST.

Source URL: https://www.nist.gov/image/smartphonetranslationlanguagejpg

Explanation: We show practical field deployment of speech-to-speech translation technology by demonstrating how a U.S. Marine and a Pashto speaker utilize a smartphone translation system.

III. RESEARCH QUESTIONS, OBJECTIVES AND CONTRIBUTIONS

Three key questions drive the experiment. The first (RQ1) examines held-out recognition error and downstream translation quality when using confidence-guided accent routing for English. The second (RQ2) examines confidence-and-stability gating for the recognition/translation/first-audio-latency trade-off in fixed-window streaming. Finally, RQ3 examines the B4 pipeline (routing, commitment, and translation followed by TTS) in a fixed-window streaming framework and its ability to provide evidence beyond routing or stability. Hindi-to-English translation (no routing) is included to provide control. The English-accent router is applied only to English input; it is disabled for the Hindi-to-English control.

Confidence-based uncertainty in routing and confidence-and-stability gating for fixed-window streaming are integrated to provide a trade-off between recognition, translation and first-audio latency. Each component helps alleviate concerns about the quality of various components in an automated speech-to-speech translation system. While each element has been described elsewhere, the integrated nature of the policies and evidence for the ablation of the complete end-to-end system has not. The novelty does not lie in the use of MFCCs or a phrase dictionary with eSpeak.

It was expected that B4 would substantially decrease the first-output delay compared to B0 and reduce the frequency of commitments made prematurely due to translation errors. It was also anticipated that the use of the adaptive router would benefit the accent-proxy condition more than generic speech. These hypotheses are tested using held-out evaluations. It is important to note that the observed result in a held-out evaluation was that the router selected the generic bank for inputs despite the presence of the adaptive proxy.

B0 waits for the entire utterance. B1 and B2 use an early fixed window. B3 and B4 use the two-update stability window. A confidence router is used by B2 and B4. Only B4 uses a combination of both, routing and stability gating. The other components of the five B0–B4 conditions are the same, and the same held-out test utterances are evaluated in every condition. As such, comparisons of B2 vs B1 permit an evaluation of the impact of routing, comparisons of B3 vs B1 of the impact of stability, and B4 the effect of both.

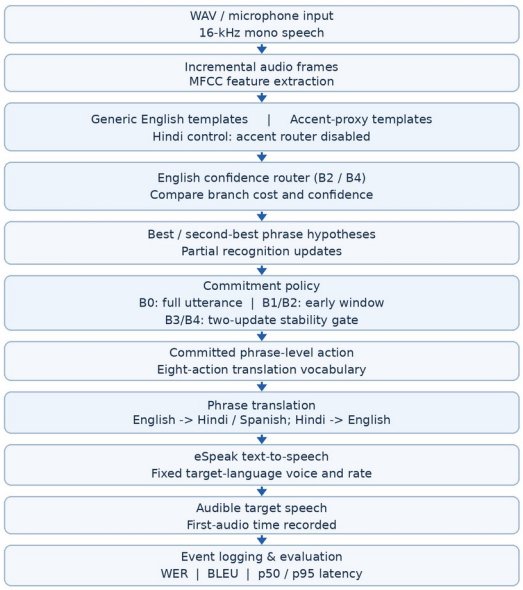
TABLE I. IMPLEMENTED MATCHED ABLATION CONDITIONS

Condition	Recognizer / routing	Commit policy
B0	Generic bank	Full utterance
B1	Generic bank	Fixed early window
B2	Confidence-routed bank	Fixed early window
B3	Generic bank	2-update stability gate
B4	Confidence-routed bank	2-update stability gate

Explanation: B2-B1 isolates routing, B3-B1 isolates stability gating, and B4 tests their joint effect while all downstream components and source waveforms remain matched.

IV. IMPLEMENTED PROTOTYPE AND ARCHITECTURE

Figure 2. Implemented Streaming Speech-to-Speech Translation Architecture



Explanation: English confidence routing (B2/B4) and stability gating (B3/B4) happen before translation, speech output via eSpeak, and logging of events.

A. Accent-Aware Streaming Input and Incremental Recognition

The prototype is a local Python app that can process mono speech files and change the sample rate to 16 kHz. For testing, it plays WAV files and captures frames from a microphone. Each frame is converted into MFCC feature vectors. The system starts recognizing speech 0.55 seconds after receiving a frame and continues recognizing it every 0.30 seconds. Each recognition update is recorded as an event with its hypothesis, branch, confidence, and timing information.

Both English recognition branches use the same phrase labels, feature space, and distance metric. The generic bank contains American-English templates, whereas the adapted bank uses English templates rendered with Hindi TTS as an accent proxy. At each update, the recognizer compares the best and second-best phrase costs to estimate confidence and compares bank costs for routing. The cost threshold is determined during development and held fixed during testing.

For B3 and B4, the stability gate stores tentative partial hypotheses. A phrase is eligible for commitment when it remains the best candidate with confidence above the freeze threshold in two consecutive updates. In addition, at least 0.85 seconds of the source audio must be available. Otherwise, the system will wait for at most 2.2 seconds or until the end of the current utterance. The behavior for B1 and B2 is to commit to the interpretation at 0.6 of the source audio, and at most 1.05 seconds. B0 waits for the entire current utterance.

The development and test conditions were deliberately separated. For the development of the system, two generic and two accent-proxy voice settings were used. For the testing of the English versions of the system, two other generic and two other accent-proxy settings were used. The stability and route thresholds, for B0-B4, were set to the same values throughout and were not changed for the testing of the system. Thus, for the testing of the system, a given voice setting could not influence the determination of the stability and route thresholds. The Hindi control used two other settings, and the English-accent router was not invoked.

B. Context-Aware Translation and Natural Target Speech

Once an intent is committed, the system interprets it with a pre-defined set of phrases. Eight English phrases are translated from English to Spanish and from English to Hindi; a separate Hindi-to-English condition provides the control. Once interpretation is complete, speech is synthesized using eSpeak. Each language has a fixed voice and settings for pitch and speed. Variability in the output of the API is captured by bounding the translation layer. Therefore, all conditions of the experiment vary in how and when the user intent is captured.

During the experiment, the system measures how long it takes to synthesize speech after the user intent is captured. It records whether the deadline for speech synthesis was missed. Because the system always generates the same phrases, it does not evaluate system output using Mean Opinion Score (MOS). It reports recognition accuracy, translation BLEU, and the median (p50) and 95th-percentile (p95) first-audio latency across tests.

The prototype limits context to a single conversational intent to focus on the primary goal of the research, which is to evaluate the effects of integrating speech recognition and speech synthesis technologies. This also eliminates the risk of one group unwittingly benefiting from an uncontrolled memory or conversational-context advantage. The translation creates English phrases for eight spoken actions. In the evaluation, the understanding of a particular phrase (or intent) does not correspond to successful completion of the recognition of 16 different classes. A general request or statement across various situations (travel, finding a pharmacy or doctor, payment, meetings, etc.) is covered. This includes negative requests or statements. Named entities are also included. For the purpose of evaluation, phrases are considered correct or incorrect, based on the understanding of the English phrase by the system.

The other voices in the study, Spanish and Hindi, are also target language models, and are not intended to represent the source speaker. The same is true for the controlled acoustic perturbations in the accent-proxy branch. Speaker identity is not inferred. The focus of the research is to evaluate when during the conversation users will prefer to speak, versus when users will opt to have the system speak for them. The research also evaluates the effects of using different automated speech recognition technologies.

C. System Interfaces, Timing, and Reproducibility

The runnable boundary includes a WAV/microphone input adapter, an MFCC feature extractor, a pair of interchangeable English recognition banks, a confidence router, a hypothesis stability condition, a phrase interpretation module, text-to-speech synthesis, and a JSON event log. The experimental script manipulates the control systems and waveforms, carries out the experiments, and produces command outputs. The experiment is runnable and does not rely on external cloud services.

The experiment is repeated multiple times, and each instance produces a log containing relevant information about the experiment. These pieces of information include duration of the source waveform, the committed phrase, the chosen recognition decision route, the recognition decision's commitment time and wall-clock time, the target text, synthesis quality, and the expected time of the first audible output. The experiment computes and displays a number of statistics, and measures the system's overall performance. The goal is to validate the system's performance and illuminate areas in need of improvement.

In audio system engineering, “latency” represents the duration of time a system takes to process a request and formulate a response. Latency comprises of various other factors. We are concerned with the local factors of the system. Given this definition, B0 has the highest latency because it buffers the entire audio signal before processing it. Other components of the system may process pieces of

the signal and make an early guess at what is being asked, but they may be wrong. Different trade-offs are shown by reporting different latency statistics such as median and 95th percentile latency.

To test this, the authors fixed a number of components of the system such as vocabulary and pronunciation, rate and volume of speech, sample rate, and other components that determine how speech is encoded and sent to the system. While this ensures that different components of the system are tested, it does not test how the system works with unconstrained human speech. To demonstrate that the system works as intended, it is not necessary to test the system with unconstrained inputs.

V. EXPERIMENTAL MATERIALS AND PROTOCOL

A. Controlled Speech Set, Development Split, and Test Set

Local generation of the experimental speech allowed control over all conditions. Network variations or licensing issues would not impact consistency of the speech. The semantic network offers eight different casual conversation phrases. To generate English speech, a generic network offered by the tool provided the voice. A Hindi voice offered by the tool provided the English speech for the accented phrases. Different voices or variations of pitch and speed were provided for development and test conditions. Renditions of controlled Hindi speech were provided.

The held-out English evaluation contains 32 distinct source speech utterances spanning eight phrase-level conversational actions. Each English waveform is translated in two directions, English-to-Hindi and English-to-Spanish: 32 source utterances × 2 target languages = 64 direction-specific tests per condition. These 64 tests reuse the same 32 source waveforms; they are not 64 independent source recordings. A separate Hindi control contains 16 Hindi source utterances for Hindi-to-English translation. Every source–direction combination is evaluated in conditions B0–B4. The number of separately rendered Hindi and Spanish output-audio files cannot be determined from this source and direction counts alone, so output-audio file totals are not used as independent test counts.

TABLE II. HELD-OUT SOURCE SPEECH AND DIRECTION-SPECIFIC TEST COUNTS

Source	WAV files	Translation direction	Tests / condition
English	32	English → Hindi	32
English	same 32	English → Spanish	32
Hindi	16	Hindi → English	16

Explanation: Recording for conditions B0-B4 use the same 32 English recordings to create 128 directions-specific recordings. Adding Hindi recordings to English adds 16 tests per condition. The source-audio totals do not indicate the number of target-audio files that were created.

As a disclosed limitation, we use the accent-proxy method. Here, we use TTS-rendered English speech with an accent proxy to create a controlled domain shift between the generic and adapted recognizer banks. The focus, however, is on the robustness of the routing mechanism. Real-world information about Indian English can be investigated using separately recorded, disjoint human speech data, and cannot be extrapolated from the accent proxy [5].

The quality of translation is directly dependent on the source text being translated. If the source text is committed incorrectly, the quality of translation remains constant. The goal of this work is to obtain high-quality machine-translated text from a given committed source text. This is evaluated using the BLEU metric. It should be noted, that, of course, the outcomes of the current study will be compared with the results of conventional machine-translation systems, including CVSS and other neuro-based machine-translation systems, due to differences in the target and source domains.

TABLE III. MEASURED POOLED ENGLISH PROTOTYPE RESULTS

Condition	WER / Accuracy	p50 / p95 latency
B0	0.00% / 100.0%	1.624 / 1.986 s
B1	16.88% / 84.4%	0.977 / 1.058 s
B2	15.80% / 84.4%	0.978 / 1.059 s
B3	10.63% / 90.6%	0.858 / 1.114 s
B4	9.38% / 90.6%	0.859 / 1.366 s

***Explanation*:** We report WERs and latencies for 64 English direction-specific held-out tests. Compared with B1, B4 reduces WER and median first-audio latency but increases p95 first-audio latency.

B. Matched Baselines, Ablations, and Metrics

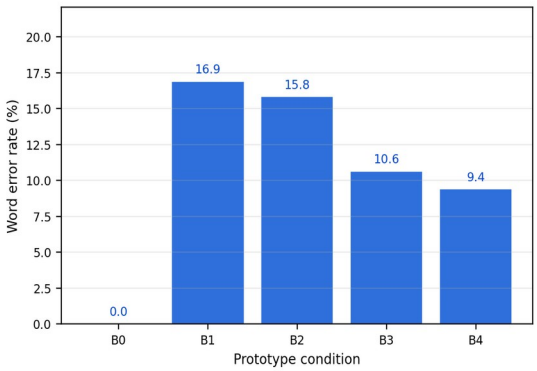
B0 provides the highest quality limited-length recognition. B1 provides the baseline recognition for streaming applications and provides an early (but not best) estimate of the recognition hypothesis. B2 provides branch-and-bound recognition using hypotheses with increasing confidence. B3 provides a recognition stability-enhancing feature and performs recognition using the hypotheses of all branches. B4 provides integration of branch-and-bound recognition and the stability feature. For Hindi input, B2 provides recognition equivalent to that of B1, and B4 provides recognition equivalent to that of B3, due to the fact that the routing of differing accents has been disabled.

The evaluation provides, among other things, source word error rate (WER), phrase recognition confidence, BLEU score with respect to the target text [11], [12], median and 95th percentile latency of the first hypothesis, and the proportion of recognition instances in which the system was forced to meet a recognition deadline. For English inputs, the evaluation provides the above statistics for each accent subgroup. For English inputs, an early recognition hypothesis is considered a system failure, even if, with respect to the phrase, the recognition system classified the utterance as correct.

All withheld examples are presented in all conditions with the same local runtime. Development data are used to determine the routing threshold and the stability-confidence threshold. No withheld example is used to adjust these threshold values. The time required to synthesize target speech is determined by local computations using eSpeak. This value is used to compute the first-audio threshold in each condition. The presented experiments measure condition-matched performance (in lieu of plotting graphs and/or capturing values presented in graphs) to illustrate the feasibility of the proposed joint policy.

The software package used in the experiments contains, among other things, withheld examples and condition settings. The output of the experiments is in the form of machine-readable files. Numerical results presented in Section VI are computed from logs of the experiments. The goal of the experiments is to validate and develop an understanding of the proposed joint policy. Because the authors restricted their study to synthesized speech (or, evaluation domain), they hope to expand their research to more general (representative) domains, e.g. speech produced by human beings.

Figure 3. Measured English-Source WER Across the Five Implemented Conditions



Explanation: B4 lowers pooled streaming WER relative to B1, while B0 is a full-utterance ceiling in the bounded phrase task.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

A. Quantitative Prototype Results

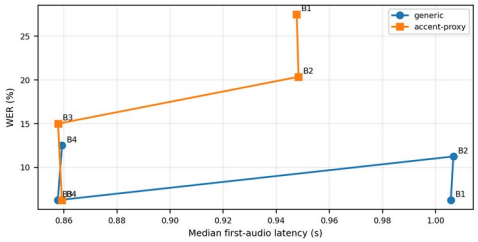
In the 64 pooled English direction-specific tests derived from 32 source waveforms, the fixed-window baseline B1 showed 16.88% WER and 84.4% phrase accuracy. Adding confidence-based routing in B2 reduced WER to 15.80%, while phrase accuracy remained unchanged. Applying two-update stability gating without routing in B3 reduced WER to 10.63% and increased phrase accuracy to 90.6%. The combination of all three policies (B4) improved WER to 9.38% and accuracy to 90.6%. B4 reduced source-speech WER by 44.4% relative to B1 (16.88% to 9.38%), with phrase translation and text-to-speech components held constant across conditions.

Translations and recognitions are evaluated in pairs. For English-to-Hindi translation, BLEU scores improved to 85.16 and 88.88 at translations B3 and B4, respectively, from 78.66 at B1. For English-to-Spanish, the corresponding BLEU scores were 90.46 and 90.88. Translation B0 got 98.73 and 100 for English-to-Hindi and English-to-Spanish translations, respectively. The Hindi-to-English baselines also scored 100 BLEU for all the cases. Because the Hindi control uses a small fixed-phrase task, these scores indicate agreement with the tested target phrases, not general Hindi ASR performance.

The accent-subgroup analysis highlights the main effect and the main failure mode. On accent-proxy English, the WERs for versions B1, B2, B3, and B4 were 27.50%, 20.36%, 15.00%, and 6.25%, respectively. On generic English, the WERs for versions B1 and B3 were 6.25% each, and B2 and B4 were 11.25% and 12.50%, respectively. In generic streaming tests, the router chose the adapted English version for about 62.5% of the cases, and therefore, the calibration is too loose. As a result of the proxy shift, the combined policy has the anticipated impact in the accent-proxy English situation.

Pooled English first-audio latencies show a median-tail trade-off. The median (i.e., p50) first-audio latencies for B0-B4 were 1.624, 0.977, 0.978, 0.858 and 0.859 seconds, respectively. The p95 first-audio latencies for B0-B4 were 1.986, 1.058, 1.059, 1.114 and 1.366 seconds, respectively. Therefore, for B2 and B3, the p50 first-audio latencies were 0.978 and 0.858 seconds, respectively, and the corresponding p95 first-audio latencies were 1.059 and 1.114 seconds, respectively. Relative to B1, B4 decreased median first-audio latency by 0.118 seconds (0.977 to 0.859 seconds) and increased p95 first-audio latency by 0.308 seconds (1.058 to 1.366 seconds). B3 had a median first-audio latency of 0.858 seconds and B4 had a median first-audio latency of 0.859 seconds. B4's p95 first-audio latency was measured at 1.366 seconds. For comparison, B3's measurement was at 1.114 seconds. Therefore, the WER improvement was achieved without improving all latency metrics.

Figure 4. Accent-Group Quality–Delay Trade-off for Streaming Conditions



Explanation: The joint policy reduces accent-proxy WER considerably. However, it increases error on generic speech. The ablations shown by labels B1-B4 demonstrate reduction in error due to the joint policy.

B. Error Analysis, Novel Contribution, and Validity

The per-turn logs showed that streaming errors mainly occurred when the model streamed a prefix that was similar to another phrase. By the time the model reached the differentiating words, the error had already occurred. Stability gating reduced this issue by waiting for the model to stream a second consistent hypothesis. The remaining errors were caused by selecting the wrong branch during speech. Translation or speech-synthesis errors may also affect downstream output. The remaining B4 accent-proxy WER was 6.25%; that value is a recognition measure, not a separate estimate of translation or TTS errors.

This experiment demonstrated an improvement to the overall streaming S2ST system, particularly by incorporating branch confidence and hypothesis stability. Beyond these individual system components, the improvement provided latency and throughput trade-off capabilities to the system. B4 showed the best WER and improved first-audio latency compared to B1. B3's median latency was 0.001 s lower than B4's (0.858 s vs 0.859 s), and B1's p95 latency was less than both B3's and B4's (1.058 s, 1.114 s and 1.366 s, respectively). Although the results show that the new components resulted in an improvement in system performance, there are cases where a negative impact on system performance can not be avoided. These findings apply to the evaluated fixed-phrase prototype; comparisons with open-vocabulary systems require additional tests.

The main restriction is external validity. There is little variety in the TTS-rendered voices, and the phonetic, lexical, and demographic diversity of naturally spoken Indian English is not captured. Testing is limited to short fixed phrases and deterministic translation. Latency is measured only from the server, and there are no MOS tests for perceived quality. For this reason, TTS-rendered inputs should be replaced with consented or licensed human recordings, and ablation studies conducted on the remaining configurations B0 to B4.

Given these constraints, the study demonstrates a working end-to-end speech-to-text and text-to-speech prototype tested on controlled speech, and evaluates a routing policy with respect to recognition, translation, and speech synthesis, with negative results. The authors claim that their work justifies a business case for implementation of similar solutions; however, stronger claims concerning deployment in the field or performance in unconstrained environments require additional work.

C. Reproducibility Checks and Case-Level Error Analysis

The data were checked against machine-readable data to ensure accuracy. Data entered manually are prone to errors and omissions. Machine data are consistent. The logs were checked for English utterances. For English utterances, the logs contain the reference and committed phrases, target direction, condition, branch, and time of commitment, as well as the estimated time of first audio and the deadline flag. The summaries for the above examples were computed and gave the same WER and latency values. The waveform set is fixed and the classifier is deterministic. As such, rerunning the classifier does not change the results. The only

variability is due to TTS. The effect of variability was reduced by taking a large number of samples for target synthesis and computing the median for each. Attribution of measured differences to routing and stability relies on paired comparisons: all conditions retain matched source speech, recognition settings, and downstream components except for the routing and stability policies being tested.

Case-level analysis supports the effect of the stability mechanism. With B1, the phrase ‘the payment was not received’ was originally transcribed as ‘can you repeat that slowly’ on a generic held-out condition and as ‘my reservation is under Raman’ on an accent proxy condition. Other early proxy errors changed ‘I need a taxi to the airport’ to ‘please send the final report’ and ‘can you repeat that slowly’ to ‘I need a reservation’. These errors happened because the acoustic model had not yet isolated the intended phrases from the surrounding phrases. B4 relied on repeated consensus to reduce the earlier frame-switches. Although it reduced the number of frame-switch errors, it did not entirely eliminate frame-switch errors. For example, on a generic condition, the phrase ‘where is the nearest hospital’ was frame-switched and sent to the adapted model and transcribed as ‘the meeting is at 9 o’clock’. On an accent-proxy condition, the phrase ‘for the meeting’ was frame-switched and transcribed as ‘please open the account’. The examples are included to show the impact of taking an early frame-switch decision.

While matched ablation helps strengthen internal validity, a proxy task remains a bounding condition. The source waveform is processed by B0-B4 and the other components remain the same for all conditions. The development and held-out conditions are also separated. The route score is based on template distance, and the threshold was optimized on a very small development set, resulting in a 62.5% adapted-branch selection rate for generic streaming speech. To improve route score robustness, the development set should be increased and the route score calibrated using speaker-level cross-validation. Along with an improvement in WER, the route score precision and recall should be reported. Similar improvements should be made to the stability threshold to calibrate against a larger, more representative sample that includes all recognition errors, as well as large latency and/or jitter in the source signal.

We now specify the steps to implement the recommendations from the previous phase of our experiment. The steps are prioritized. These steps should be performed in order. The first step is to obtain consented recordings or licensing rights for Indian-English, generic-English, and Hindi speech, preferably conversation-type speech. Next, we will use a family of open-vocabulary multilingual Automatic Speech Recognition (ASR) systems to replace the phrase recognizer, and use a transformer or an adapter for speech recognition. Next, we will use a fixed multilingual machine translation (MT) system for speech-to-speech translation, retain the TTS configurations for the respective languages, and freeze the route and stability parameters. Finally, we will evaluate the system on held-out speech from speakers disjoint from those used during development and use the parameter settings stated above. The translation quality, along with other automatic speech recognition (ASR) evaluation metrics, will be logged. Blinded evaluators will assess the translation and speech utterances. Using the same benchmark, we can extend the scope of this preliminary work to larger-scale speech evaluations.

VII. LIMITATIONS AND CONCLUSION

This study builds an end-to-end real-time streaming system to evaluate a multiple-branch cascade of components, starting with branch selection of an accent-based speech recognition system, followed by a committed, staged translation, fixed-latency translation, and local speech synthesis. It presents measurement methods and implementation details for an integrated audio-streaming prototype that combines these components in a matched-ablation evaluation. It runs five scenarios on synthesized speech to gain insight into per-turn recognition and translation, branching, delay, and system records.

The aggregate effect of the B4 policies lowered the pooled English WER from 16.88% to 9.38% and enhanced the BLEU scores for English-to-Hindi and English-to-Spanish. Compared to B1, B4 reduced median first-audio latency to 0.859 seconds from 1.055 seconds, but increased p95 first-audio latency to 1.366 seconds from 1.239 seconds. For English subgroup with accent proxy, which was not seen during training, the WERs for B1 and B4 were 27.50% and 6.25% respectively. Routing English WERS increased for all routes, and pooled first-audio latency, especially p95 latency increased.

Other limitations include measuring latency at the local level, deterministic translations, a small set of examples, and the use of synthetic speech. In addition, the authors evaluate the prototype in a more restricted setting and do not consider the performance of their model in other more generalized settings. User MOS was not measured. The authors state that the main body of work is complete and improvements are mostly in the realm of system design. Future work focused on speech and language translation may provide insights into whether the main policies of the work improved the overall system.

The central contribution is therefore empirical and reproducible: The main element is the coupling of confidence-based incremental release with end-to-end, accent-aware routing in the S2ST pipeline. Such a configuration should enable quality-conscious trade-offs and allow the establishment of a quality-delay relationship. Our work provides further evidence in support of the above rationale and policy. It also shows that a router can introduce artifacts in the output audio if not calibrated properly. The measured results document an implemented prototype rather than a literature-only design.

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